IRWA Final Project – Part 1: Text Processing and Exploratory Data Analysis

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GitHub https://github.com/victorbrull/irwa-search-engine
Tag `IRWA-2026-part-1`

1. Introduction

The corpus contains **28,080 fashion products** crawled from Flipkart, with 17 fields each. In this first part we prepare the documents, and we describe the data with an exploratory analysis. Before deciding anything we inspected the raw data, and several of our decisions come from data-quality problems we found there (section 2). For the decisions about the fields (questions 3 and 4) we used `validation_labels.csv` as evidence, as suggested in the statement.

All the code is in the notebook `processing_eda.ipynb`. The pre-processing functions are in a separate python file, `/preprocessing.py`, to be cleaner.

2. Raw data and quality problems

`pid` is unique (no duplicates), so we use it as the document identifier. Missing values are stored as empty strings, not `null`. The most important ones are: **description empty in 39.7%** of the products, rating missing in 8.1%, brand in 7.2%, seller in 5.9% and actual price/discount in ~2.8%. Prices, discount and rating are strings (`"2,999"`, `"69% off"`, `"3.9"`) and `product_details` is a list of one-key dictionaries. While inspecting the data we found four problems that affect retrieval:

1. **Encoding errors.** 6.5% of descriptions contain non-ASCII characters (curly quotes, non-breaking spaces) and 43 have real *mojibake* (text decoded with the wrong encoding, sometimes twice, e.g. `itÃ\x83Â¢Ã\x82Â\x80…s` instead of *it's*).
2. **Glued words.** In 7.4% of descriptions, sentences are joined without a space (`look.Designed`, `Comfort,Skin`) and sometimes even the brand is glued to the next word (`ECKO UnltdSlim Fit`).
3. **Truncated brands.** Most brand names are cut: `Pu`, `REEB`, `ECKO Unl`, `True Bl`. A user searching for *puma* would never match `pu`. The URL slug, however, starts with the complete brand name (`flipkart.com/puma-full-sleeve-…`), so we use it to complete the brand. This works for 24,741 of the 26,071 products with a brand (94.9%).
4. **Gender in the URL contradicts the title.** We compared the gender word (*men*/*women*) in the title, the URL and the `Ideal For` detail. Title and `Ideal For` agree in **100%** of the 16,433 products where both exist, but the URL agrees with them only **~50%** of the time, which is what you would get by chance. In the validation file all 20 products of query 1 say *Women* in the title and *men* in the URL, and the labels follow the title. Our conclusion is that the visible fields were modified and the URL was not, so **we never index the URL as text**. We only use it to complete the brand and to give the user a link.

3. Text pre-processing

The function `preprocess_text` is applied to every text field and will also be applied to queries:

#	Step	Reason / detail
1	Encoding repair (`ftfy`)	fixes the mojibake of problem 1
2	ASCII folding (Unicode NFKD)	`’` → `'`, `é` → `e`, removes symbols such as `•` and `™`
3	Split glued words	`look.Designed` → `look. Designed`; `UnltdSlim` → `Unltd Slim` (problem 2)
4	Lowercase	*Cotton* = *cotton*
5	Domain normalisation	`women's` / `womens` / `woman` → `women`; `t-shirt` / `t shirt` / `tee` → `tshirt`; `sweat shirt` → `sweatshirt`; typo `regulor` → `regular`
6	**Tokenization** with regex `[a-z0-9]+`	every other character is a separator, so **punctuation is removed** in the same step
7	**Stop-word removal** (NLTK English)	we checked that the list does not contain *men, women, full, slim, blue*, which are key query terms
8	Remove 1-char tokens and product codes	codes that mix letters and digits (`1005combo2`) are identifiers; pure numbers (*100*% cotton, size *32*) are kept
9	**Stemming** (Porter)	`sleeves` → `sleev`, `jeans` → `jean`; Porter keeps `sleeveless` different from `sleev`, which is what we want

Steps 1–3, 5 and 8 are our additions (the "anything else" point), and each one fixes a problem seen in the data. We used a regex tokenizer instead of `nltk.word_tokenize` because `word_tokenize` keeps punctuation as tokens and handles hyphens inconsistently. For example, the two validation queries become `[women, full, sleev, sweatshirt, cotton]` and `[men, slim, jean, blue]`. To measure the effect of each step, we computed the vocabulary of title + description after adding the steps one by one:

Pipeline	Vocabulary	Avg. tokens / doc
whitespace split	13,521	36.2
+ lowercase	10,606	36.2
+ punctuation removal (regex tokenizer)	6,503	37.6
+ encoding fix, glued words, domain normalisation	6,432	36.7
+ stop words, codes, 1-char tokens	6,187	25.0
+ Porter stemming (final)	4,717	25.0

Lowercasing and punctuation removal halve the vocabulary, and stemming reduces it by another 24%. Stop words barely change the vocabulary, but they remove about a third of the tokens of each document. The normalisation step has a small effect on size (−71 terms), but it fixes exactly the cases that matter for matching (`women's`, `t-shirt`, `UnltdSlim`).

4. Fields returned to the user

For every document the processed corpus keeps the 14 required fields (`pid`, `title`, `description`, `brand`, `category`, `sub_category`, `product_details`, `seller`, `out_of_stock`, `selling_price`, `discount`, `actual_price`, `average_rating`, `url`) in a **readable, typed form**: original text with only the encoding fixed, prices as numbers, `product_details` as a dictionary. The stemmed terms are stored in separate columns (`<field>_terms`) because they are only used for matching. Missing values are returned as `None` instead of `""` or 0, so the interface can show "not available". The original (truncated) brand is shown to the user, and the completed brand (`brand_full`) is only used for search.

5. Q3 – category, sub_category, brand, product_details and seller

Option	Pros	Cons
A. Merge all into one text field	simplest index; any term matches anywhere; works directly with TF-IDF + cosine	all fields weigh the same (a match in a washing instruction counts like one in the title); repeated values make documents longer; no filtering by brand/seller
B. Separate fields in the index	each field gets its own weight (title > details > description); field queries and filters (`brand:puma`); per-field statistics	more complex index and ranking; weights must be tuned; very short fields (brand) have skewed length statistics
C. Structured attributes, not text	low-cardinality fields work better as filters or facets	they don't help free-text matching

To decide, we looked at (1) how distinctive each field is and (2) whether it helps separate relevant from non-relevant products in the validation set.

Field	Distinct values	Missing	Most common value	Entropy (bits)
category	4	0%	Clothing and Accessories (96.6%)	0.22
sub_category	24	0%	Topwear (54.5%)	2.28
brand (completed)	323	7.2%	ecko unltd (3.6%)	6.83
seller	534	5.9%	RetailNet (5.3%)	7.13

`category` carries almost no information: as a term its IDF is ≈ 0. `brand` and `seller` are very distinctive, but users type brands in queries, not seller names. `product_details` contains exactly the attributes people search for: *Sleeve* (Full Sleeve in 6,372 products), *Fit* (Slim in 3,831), *Fabric*, *Color*, *Ideal For*. Then, for each validation product, we computed the fraction of the query terms found in each field (Figure 1).

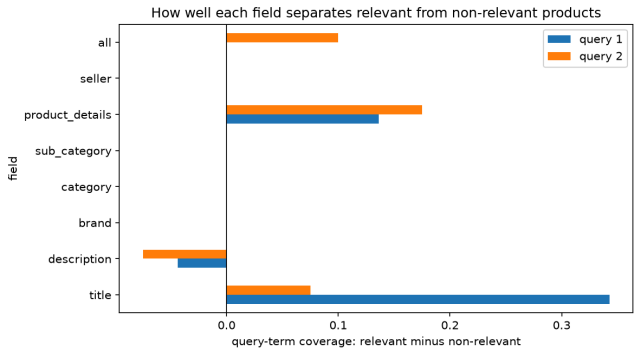


Figure 1. Query-term coverage of relevant minus non-relevant products, per field.

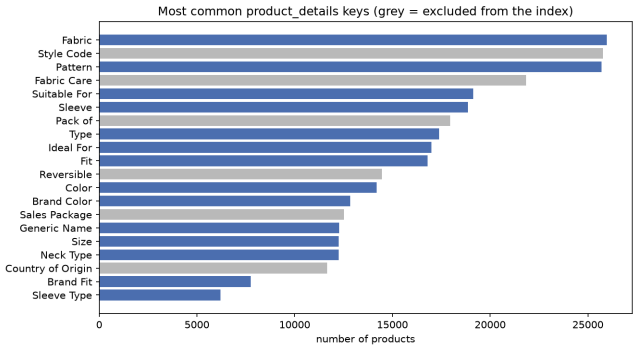


Figure 2. Most common product_details keys (grey = excluded).

The result was clear. With a **single merged field**, all the candidates of query 1 (relevant or not) contain the five query terms somewhere, so the coverage is 1.00 vs 1.00 and the field cannot separate them. The **title** separates best (0.80 vs 0.46 for query 1), because the non-relevant products are T-shirts that mention *sweatshirt* or *cotton* only in other fields. **product_details** also helps (+0.14 / +0.18): the *Sleeve* value separates all four T-shirts (*Half Sleeve*) of query 1, and *Fabric* explains one of the non-relevant sweatshirts (*Polyester Blend*). The **description** is slightly negative, and brand, category, sub_category and seller never contain the query terms.

> **Decision (a hybrid of A, B and C).**
>
> 1. We keep a **separate term list per field** (title, description, product_details, brand, sub_category, category, seller), so that in Part 3 we can weight them (title > product_details > brand > description) or use a field-weighted model such as BM25F.

2. We also build a **merged field** `all_terms` = title + description + brand + sub_category + product_details for the basic TF-IDF index of Part 2 (simple, good recall). `category` (no discriminative power) and `seller` (noise for product queries) are **not** merged.
3. `category`, `sub_category`, `brand` and `seller` are also kept as **raw values for filters/facets**.
4. From `product_details` we index **only the values** (keys such as *Fabric* appear in almost every product), and we drop keys that don't describe the product: *Style Code, Fabric Care, Sales Package, Country of Origin, Pack of, Reversible, Number of Contents in Sales Package*.

Note about the labels. They are partly inconsistent. Products with the same title and brand get different labels (Puma *Full Sleeve Printed Women Sweatshirt*: 1, 1, 0; ECKO *Tapered Fit Men Blue Jeans*: 0, 1, 1). Two cotton-blend Puma sweatshirts are non-relevant although they look like the relevant ones, and three *Women* jeans are relevant for *men slim jeans blue*. No text-only ranking can reproduce these labels perfectly, and we will keep that in mind when we evaluate in Part 2.

6. Q4 – out_of_stock, selling_price, discount, actual_price, average_rating

They are not indexed as textual terms. As terms, *921* and *920* would be unrelated, while what matters about a number is its order and distance (cheaper than, rated above), which an inverted index of terms cannot express. Users ask for *"under 500"* or *"best rated"*, which are range filters or ranking signals, not exact matches. Number tokens would also add thousands of useless terms. Instead, we parse them into typed attributes: `"2,999"` → 2999.0, `"69% off"` → 69.0, `"3.9"` → 3.9, `out_of_stock` → boolean. **Missing values stay missing (never 0)**: an unrated product is not a product rated 0, and a missing price does not mean free. If the discount is missing but both prices exist, we compute it. We checked the consistency of the fields: the stated discount matches `(actual - selling)/actual` within one point for 100% of the products, and no product has a selling price above its actual price. For the next parts we plan to use `out_of_stock` as a filter or penalty, `selling_price` as a range filter and for sorting, and `discount` and `average_rating` as ranking signals combined with the text score.

7. Exploratory data analysis

7.1 Text statistics

Titles are short and very regular: 6.5 words on average (median 6). Descriptions are empty in 39.7% of products, and the non-empty ones have 49.4 words on average (median 39, max 623), with 3.2 sentences of 15.3 words. Sentence splitting was done after repairing the glued sentences, otherwise `look.Designed` would not be split. After pre-processing, the collection has **1.26M terms and a vocabulary of 5,865 terms** (1,216 appear only once). Per field, the vocabulary is 716 (title), 4,528 (description) and 2,960 (product_details).

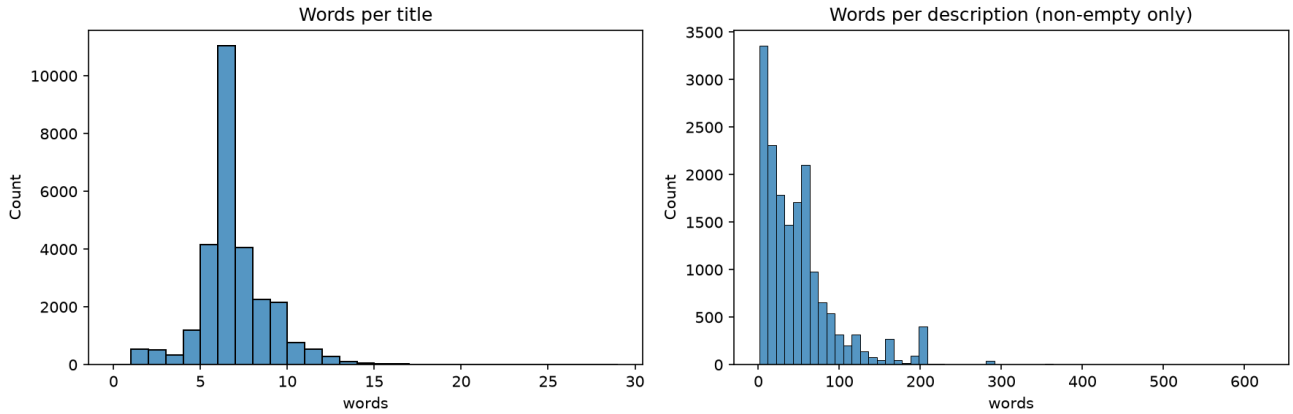


Figure 3. Words per title and per non-empty description.

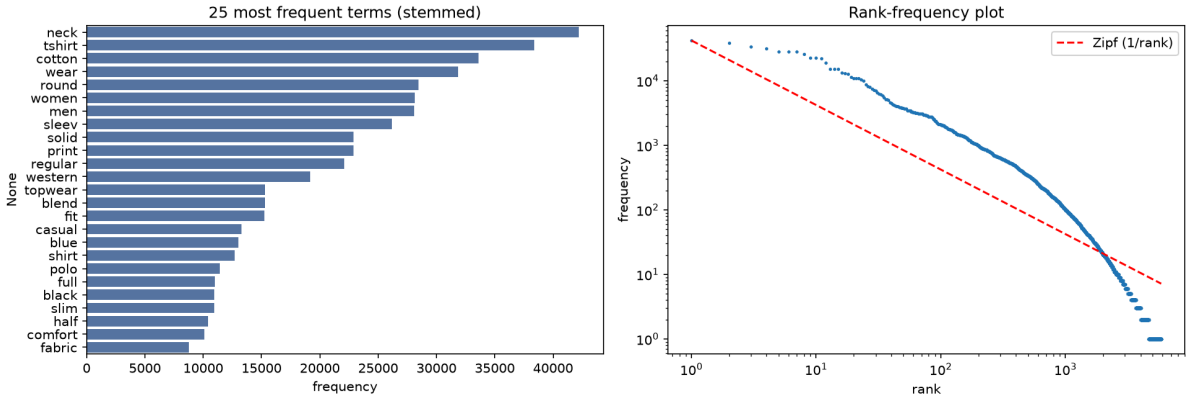


Figure 4. Most frequent terms and rank-frequency plot (log-log) compared with Zipf's law.

The rank-frequency curve follows Zipf's law in the middle ranks. The most frequent terms (*neck, tshirt, cotton, wear, round, women, men, sleev*) come largely from `product_details`. For example, *Suitable For: Western Wear* appears in 19k products. This matters for the validation queries: *cotton* is in 71% of the documents and *sleev* in 64%, so their IDF is low, and *sweatshirt* (5%) and *jean* (9%) will do most of the ranking work.



Figure 5. Word clouds of titles and descriptions (pre-processed without stemming so that the words are readable).

Titles are dominated by gender and product type (*women, men, tshirt, neck, printed, solid*), while descriptions use marketing vocabulary (*wear, casual, look, comfort, quality, perfect*). This is one more argument for giving more weight to the title.

7.2 Categories, brands and sellers

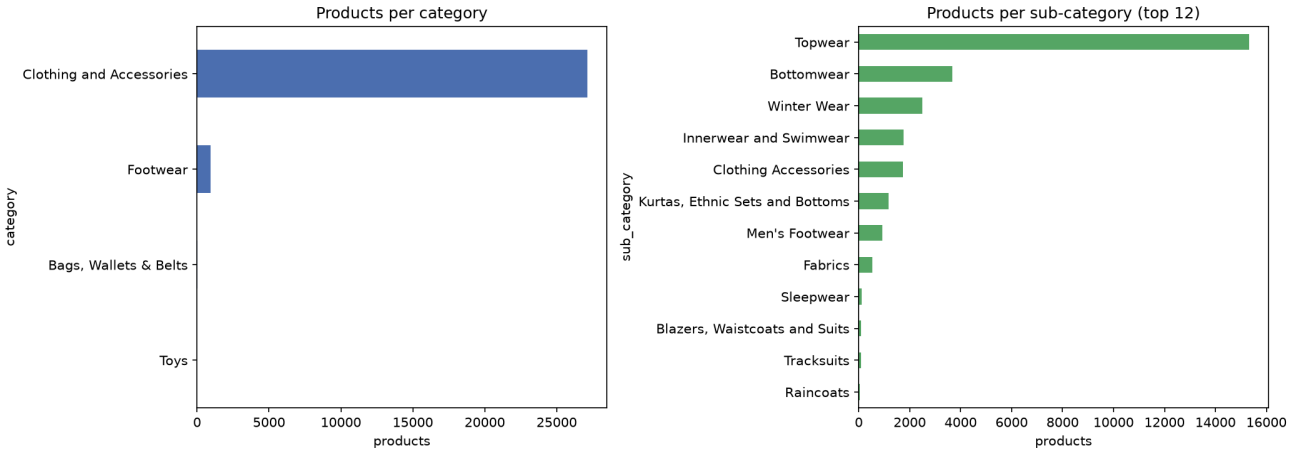


Figure 6. Products per category and sub-category.

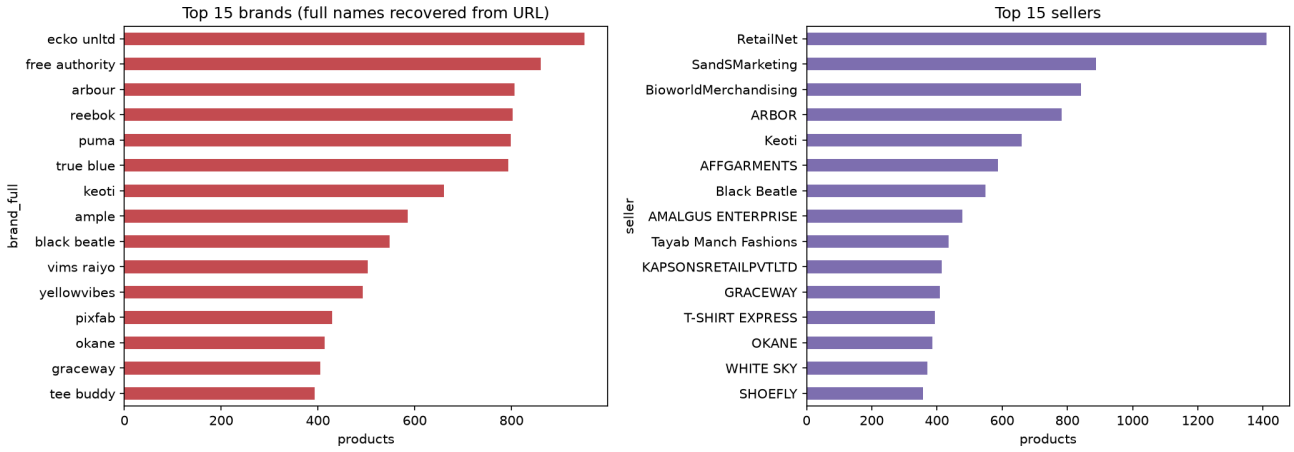


Figure 7. Top 15 brands (completed from the URL) and sellers.

The collection is almost only clothing (96.6%), and more than half is *Topwear*. There are 323 brands and 534 sellers. The largest brands are ECKO Unltd (951), Free Authority (860), Arbour (806), Reebok (802) and Puma (798). Without completing the brands, the chart would show *ECKO Unl*, *Free Authori* or *Pu*. Among the brands with at least 50 rated products, the best average ratings are Sorang (4.61), PixFab (4.46) and Cotson (4.44); among sellers, SORANG (4.61), EVINCEMARKETERS (4.48) and PixFab (4.46). RetailNet, the largest seller (1,411 products), has a good average of 4.11.

7.3 Stock, prices, discounts and ratings

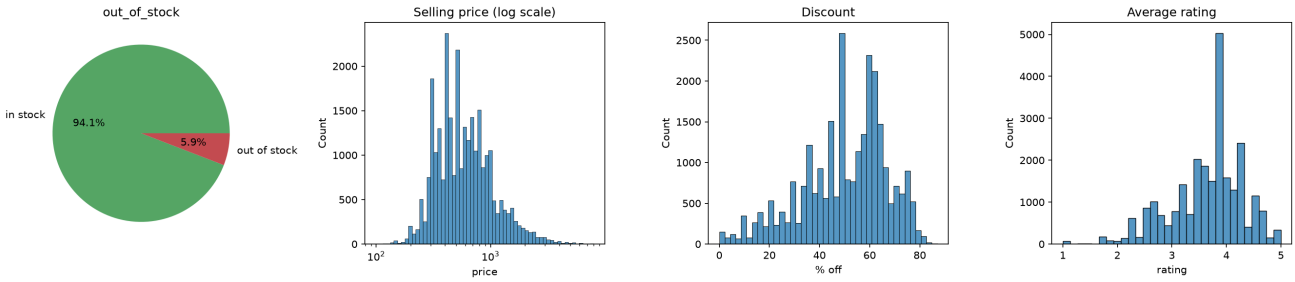


Figure 8. out_of_stock, selling price (log scale), discount and rating distributions.

5.9% of the products are out of stock (1,644). Selling prices range from 99 to 7,999 (median 545), and the actual price has a median of 1,199. Discounts are very high (median 52%, maximum 87%, 0% for some products). Ratings go from 1.0 to 5.0 (median 3.8). **329 products have a perfect 5.0** (several out of stock), and since we don't have the number of reviews, these are probably based on few ratings. Ranking only by rating would push them to the top, so in Part 3 we will combine the rating with the text score instead. Among the 10 largest sub-categories, *Clothing Accessories* has the highest out-of-stock rate (17%) and the best mean rating (3.95), and *Blazers, Waistcoats and Suits* is the most expensive (median 1,799).

8. Conclusions and next steps

The main lesson of this part was that the data needed more care than the standard pipeline: encoding repair, glued words, truncated brands, and a URL that contradicts the title. The validation labels showed that a single merged field cannot separate relevant from non-relevant products, while the title and the product details can. For this reason we keep a term list per field plus a merged field, and keep the numeric fields as typed attributes. The processed corpus is saved in `data/processed_corpus.pkl` and will be the input of the inverted index and the TF-IDF ranking of Part 2.

AI usage statement

We used Claude (Anthropic) as an assistant during this part: to help write and debug the pre-processing functions and help in the notebook code, to suggest checks on the data (e.g. the comparison between title, URL and *Ideal For*), and to draft parts of this report. We ran all the code ourselves, checked that every number in the report matches the notebook output, and reviewed and edited the text. The decisions and their justification are our own responsibility.